Maneuvering Control Problems for a Spacecraft with Unactuated Fuel Slosh Dynamics

Mahmut Reyhanoglu

Abstract— **This paper studies maneuvering control problems for a spacecraft with fuel slosh in a zero gravity environment. The spacecraft is represented as a rigid body and fuel slosh dynamics are included using a single vibrating mass model. Both pendulum and mass-spring analogies are considered. The control inputs are defined by a thrust along the spacecraft's longitudinal axis, a transverse body fixed force and a pitching moment about the center of mass of the spacecraft. The control objective is to suppress the slosh mode, as well as the transverse and pitch dynamics, while the spacecraft accelerates in axial direction. A nonlinear feedback controller is developed to achieve this objective.**

Index Terms— **Nonlinear feedback control, spacecraft, fuel slosh dynamics, underactuated mechanical system.**

I. Introduction

WITH today's large and complex spacecraft, a substantial mass of fuel is necessary to place them into orbit and to perform orbital maneuvers. The mass of fuel contained in the tanks of a geosynchronous satellite amounts to approximately 40% of its total mass [12]. When the fuel tanks are only partially filled, large quantities of fuel move inside the tanks under translational and rotational accelerations and generate the slosh dynamics. Several methods have been employed to reduce the effect of sloshing, such as introducing baffles inside the tanks or dividing a large container into a number of smaller ones. These techniques, although helpful in some cases, do not completely succeed in cancelling the sloshing effects.

It is well known that during thrust maneuvers the sloshing of fuel in partially filled tanks can interact with the controlled system in such a way as to cause the overall system to be unstable. Hence, the slosh dynamics should be included in the dynamic model of the spacecraft used for control synthesis. There is an extensive literature (see e.g. [2], [7], [12], [13]) on the interaction of spacecraft dynamics and slosh dynamics and their control, but this literature treats only the case of small perturbations to the spacecraft dynamics rather than the nonlinear translational and rotational spacecraft dynamics that are treated in this paper.

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It has been demonstrated [7] that pendulum and mass-spring models can approximate complicated fluid and structural dynamics; such models have formed the basis for many studies on dynamics and control of space vehicles with fuel slosh. For accelerating space vehicles, several thrust vector control design approaches have been developed to suppress the fuel slosh dynamics. These approaches have commonly employed methods of linear control design ([2], [12], [13]) and adaptive control [1]. A related paper, following a similar approach, is motivated by a robot arm moving a liquid filled container [5]. In most of these approaches, suppression of the slosh dynamics inevitably leads to excitation of the transverse vehicle motion through coupling effects; this is a major drawback which has not been adequately addressed in the published literature.

In this paper a spacecraft with a partially filled spherical fuel tank is considered and the lowest frequency slosh mode is included in the dynamic model using pendulum and mass-spring analogies. A complete set of spacecraft control forces and moments is assumed to be available to accomplish planar maneuvers. Aerodynamic effects are ignored here, although they can be easily included in the spacecraft dynamics assuming that they are cancelled by the spacecraft controls. It is also assumed that the spacecraft is in a zero gravity environment, but this assumption is for convenience only. These simplifying assumptions render the problem tractable, while still reflecting the important coupling between the unactuated slosh dynamics and the actuated rigid body motion of the spacecraft. The control objective, as is typical for spacecraft orbital maneuvering problems, is to control the translational velocity vector and the attitude of the spacecraft, while attenuating the slosh mode. Subsequently, mathematical models that reflect all of these assumptions are constructed. Finally, a nonlinear feedback control law is designed to achieve this control objective.

II. Model Formulation

In this section, we formulate the dynamics of a spacecraft with a spherical fuel tank and include the lowest frequency slosh mode. We represent the spacecraft as a rigid body (base body) and the sloshing fuel mass as an internal body, and follow the development in our

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previous work [3] to express the equations of motion in terms of the spacecraft translational velocity vector, the angular velocity, and the internal (shape) coordinate representing the slosh mode.

To summarize the formulation in [3], let $v \in \mathbf{R}^3$, $\omega \in \mathbf{R}^3$, and $\xi \in \mathbf{R}$ denote the base body translational velocity vector, the base body angular velocity vector, and the internal coordinate, respectively. In these coordinates, the Lagrangian has the form $L = L(v, \omega, \xi, \dot{\xi})$, which is $SE(3)$-invariant in the sense that it does not depend on the base body position and attitude. The generalized forces and moments on the spacecraft are assumed to consist of control inputs which can be partitioned into two parts: $\tau_t \in \mathbf{R}^{n_t}$ (typically from thrusters) is the vector of generalized control forces that act on the base body and $\tau_r \in \mathbf{R}^{n_r}$ (typically from symmetric rotors, reaction wheels, and thrusters) is the vector of generalized control torques that act on the base body. We also assume that the internal dissipative forces are derivable from a Rayleigh dissipation function R. Then, the equations of motion of the spacecraft with internal dynamics are shown to be given by:

$$\frac{d}{dt}\frac{\partial L}{\partial v} + \hat{\omega}\frac{\partial L}{\partial v} = \tau_t , \tag{1}$$

$$\frac{d}{dt}\frac{\partial L}{\partial \omega} + \hat{\omega}\frac{\partial L}{\partial \omega} + \hat{v}\frac{\partial L}{\partial v} = \tau_r , \tag{2}$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\xi}} - \frac{\partial L}{\partial \xi} + \frac{\partial R}{\partial \dot{\xi}} = 0 , \tag{3}$$

where $\hat{a}$ denotes a 3×3 skew-symmetric matrix formed from $a = (a_1, a_2, a_3) \in \mathbf{R}^3$:

$$\hat{a} = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} .$$

Note that equations (1)-(2) are identical to Kirchhoff's equations [6], which can also be expressed in the form of Euler-Poincaré equations. It must be pointed out that in the above formulation it is assumed that no control forces or torques exist that directly control the internal dynamics. The objective is to simultaneously control the rigid body dynamics and the internal dynamics using only control effectors that act on the rigid body; the control of internal dynamics must be achieved through the system coupling. In this regard, equations (1)-(3) model interesting examples of underactuated mechanical systems. In our previous research ([8], [9]), we have developed theoretical controllability and stabilizability results for a large class of underactuated mechanical systems using tools from nonlinear control theory. We have also developed effective nonlinear control design methodologies [11] that we applied to several examples of underactuated mechanical systems, including underactuated space vehicles ([4], [10]).

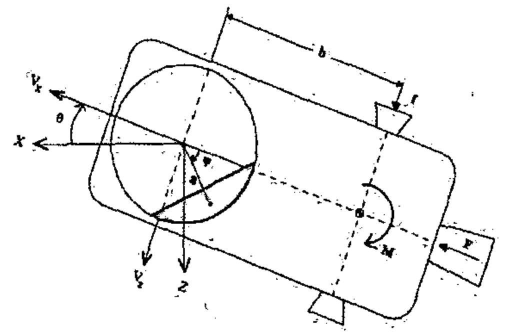

Fig. 1. Model of a Spacecraft with Slosh Dynamics: Pendulum Analogy.

A. Pendulum Analogy

The formulation using pendulum analogy can be summarized as follows. Consider a rigid spacecraft moving in a fixed plane as indicated in Figure 1. The important variables are the axial and transverse components of the velocity vector, v_x, v_z, the attitude angle θ of the spacecraft with respect to a fixed reference, and the angle ψ of the pendulum with respect to the spacecraft longitudinal axis, representing the fuel slosh. A thrust F, which is assumed to act through the spacecraft center of mass along the spacecraft's longitudinal axis, a transverse force f, and a pitching moment M are available for control purposes. The constants in the problem are the spacecraft mass m and moment of inertia I (without fuel), the fuel mass m_f and moment of inertia I_f (assumed constant), the length $a > 0$ of the pendulum, and the distance b between the pendulum point of attachment and the spacecraft center of mass location along the longitudinal axis; if the pendulum point of attachment is in front of the spacecraft center of mass then $b > 0$. The parameters m_f, I_f and a depend on the shape of the fuel tank, the characteristics of the fuel and the fill ratio of the fuel tank.

Under the indicated assumptions, the total kinetic energy is given by

$$\begin{aligned} T = {} & \frac{1}{2}m[v_x^2 + (v_z + b\dot{\theta})^2] + \frac{1}{2}I\dot{\theta}^2 + \frac{1}{2}I_f(\dot{\theta} + \dot{\psi})^2 \\ & + \frac{1}{2}m_f[(v_x + a(\dot{\theta} + \dot{\psi})\sin\psi)^2 \\ & + (v_z + a(\dot{\theta} + \dot{\psi})\cos\psi)^2] . \end{aligned}$$

Since gravitational effects are ignored, there is no potential energy. Thus, the Lagrangian equals the kinetic energy, i.e. $L = T$. Applying equations (1)-(3) with

$$\xi = \psi,\ R = \frac{1}{2}\epsilon\dot{\psi}^2,\ v = \begin{pmatrix} v_x \\ 0 \\ v_z \end{pmatrix},\ \omega = \begin{pmatrix} 0 \\ \dot{\theta} \\ 0 \end{pmatrix},$$

$$\tau_t = \begin{pmatrix} F \\ 0 \\ f \end{pmatrix}, \ \tau_r = \begin{pmatrix} 0 \\ M + fb \\ 0 \end{pmatrix},$$

the equations of motion can be obtained as

$$(m + m_f)(\dot{v}_x + \dot{\theta} v_z) + m_f a(\ddot{\theta} + \ddot{\psi}) \sin\psi + mb\dot{\theta}^2 + m_f a(\dot{\theta} + \dot{\psi})^2 \cos\psi = F, \quad (4)$$

$$(m + m_f)(\dot{v}_z - \dot{\theta} v_x) + m_f a(\ddot{\theta} + \ddot{\psi}) \cos\psi + mb\ddot{\theta} - m_f a(\dot{\theta} + \dot{\psi})^2 \sin\psi = f, \quad (5)$$

$$(I + mb^2)\ddot{\theta} + mb(\dot{v}_z - \dot{\theta} v_x) - \epsilon\dot{\psi} = M + bf, \quad (6)$$

$$(I_f + m_f a^2)(\ddot{\theta} + \ddot{\psi}) + m_f a(\dot{v}_x + \dot{\theta} v_z) \sin\psi + m_f a(\dot{v}_z - \dot{\theta} v_x) \cos\psi + \epsilon\dot{\psi} = 0. \quad (7)$$

The control objective is to design feedback controllers so that the controlled spacecraft accomplishes a given planar maneuver, that is a change in the translational velocity vector and the attitude of the spacecraft, while attenuating the slosh mode.

B. Mass-Spring Analogy

The mass-spring analogy is related to the pendulum analogy, in which the oscillation frequency of the mass-spring element represents the lowest frequency sloshing mode [12].

Consider a rigid spacecraft moving on a plane as indicated in Figure 2, where v_x, v_z are the axial and transverse components of the velocity vector and θ denotes the attitude angle of the spacecraft with respect to a fixed reference. The slosh mode is modeled by a point mass m_f whose relative position along the body z-axis is denoted by s; a restoring force $-ks$ acts on the mass whenever the mass is displaced from its neutral position $s = 0$. As in the previous model, a thrust F, which is assumed to act through the spacecraft center of mass along the spacecraft's longitudinal axis, a transverse force f, and a pitching moment M are available for control purposes. The constants in the problem are the spacecraft mass m and moment of inertia I, the fuel mass m_f, and the distance b between the body z-axis and the spacecraft center of mass location along the longitudinal axis. The parameters m_f, k and b depend on the shape of the fuel tank, the characteristics of the fuel and the fill ratio of the fuel tank.

Under the indicated assumptions, the Lagrangian is given by

$$\begin{aligned} L &= \frac{1}{2}m[(v_x^2 + (v_z + b\dot{\theta})^2] + \frac{1}{2}I\dot{\theta}^2 - \frac{1}{2}ks^2 \\ &+ \frac{1}{2}m_f[(v_x + s\dot{\theta})^2 + (v_z + \dot{s})^2] \end{aligned}$$

Applying equations (1)-(3) with

$$\xi = s, \ R = \frac{1}{2}c\dot{s}^2, \ v = \begin{pmatrix} v_x \\ 0 \\ v_z \end{pmatrix}, \ \omega = \begin{pmatrix} 0 \\ \dot{\theta} \\ 0 \end{pmatrix},$$

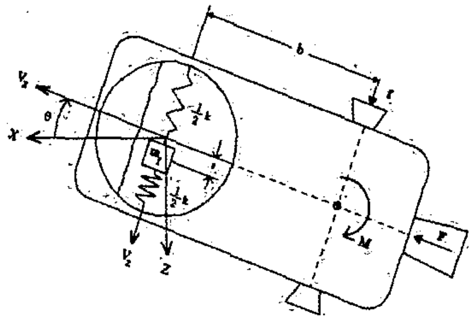

Fig. 2. Model of a Spacecraft with Slosh Dynamics: Mass-Spring Analogy.

$$\tau_t = \begin{pmatrix} F \\ 0 \\ f \end{pmatrix}, \ \tau_r = \begin{pmatrix} 0 \\ M + fb \\ 0 \end{pmatrix},$$

the equations of motion can be obtained as

$$(m + m_f)(\dot{v}_x + \dot{\theta} v_z) + m_f s\ddot{\theta} + mb\dot{\theta}^2 + 2m_f\dot{s}\dot{\theta} = F, \quad (8)$$

$$(m + m_f)(\dot{v}_z - \dot{\theta} v_x) + mb\ddot{\theta} + m_f\ddot{s} - m_f s\dot{\theta}^2 = f, \quad (9)$$

$$(I + mb^2 + m_f s^2)\ddot{\theta} + m_f s(\dot{v}_x + \dot{\theta} v_z) + 2m_f s\dot{s}\dot{\theta} + mb(\dot{v}_z - \dot{\theta} v_x) = M + bf, \quad (10)$$

$$m_f(\ddot{s} + \dot{v}_z - \dot{\theta} v_x - s\dot{\theta}^2) + ks + c\dot{s} = 0. \quad (11)$$

The control objective is again to design feedback controllers so that the controlled spacecraft accomplishes a given planar maneuver, that is a change in the translational velocity vector and the attitude of the spacecraft, while suppressing the slosh mode.

III. Feedback Control Design

In this section, we restrict the development to the model obtained via the pendulum analogy, i.e. we design a feedback controller for the system (4)-(7) only. In our previous work [4], we considered a constant thrust $F > 0$ and designed a feedback law (using a backstepping approach) to stabilize the system to a relative equilibrium defined by a constant acceleration in the axial direction. Here, we will follow a slightly different approach and design a relatively simpler feedback controller using a Lyapunov method without resorting to backstepping.

Consider the system (4)-(7). If the axial thrust F is a positive constant and if the transverse force and pitching moment are zero, $f = M = 0$, then the spacecraft and fuel slosh dynamics have a relative equilibrium defined by

$$v_x^*(t) = \frac{F}{m + m_f}t + v_{x0}, \ v_z = v_z^*,$$

$$\theta = \theta^*, \, \dot{\theta} = 0, \, \psi = 0, \, \dot{\psi} = 0 \, ,$$

where v_z^* and θ^* are arbitrary constants, and v_{x0} is the initial axial velocity of the spacecraft. Without loss of generality, in our subsequent analysis, we let $v_z^* = 0, \, \theta^* = 0$.

As in [4], we assume that the pitch and slosh dynamics have only a negligible effect on the axial acceleration and substitute the approximation

$$\dot{v}_x + \dot{\theta} v_z = \frac{F}{m + m_f} \tag{12}$$

into equation (7). This leads to the following equations of motion for the transverse, pitch and slosh dynamics:

$$(m + m_f)(\dot{v}_z - \dot{\theta} v_x(t)) + m_f a(\ddot{\theta} + \ddot{\psi}) \cos \psi + mb\ddot{\theta} - m_f a(\dot{\theta} + \dot{\psi})^2 \sin \psi = f \, , \tag{13}$$

$$(I + mb^2)\ddot{\theta} + mb(\dot{v}_z - \dot{\theta} v_x(t)) - \epsilon \dot{\psi} = M + bf \, , \tag{14}$$

$$(I_f + m_f a^2)(\ddot{\theta} + \ddot{\psi}) + m_f a \frac{F}{m + m_f} \sin \psi + m_f a(\dot{v}_z - \dot{\theta} v_x(t)) \cos \psi + \epsilon \dot{\psi} = 0 \, , \tag{15}$$

where $v_x(t)$ is considered as an exogenous input.

Define the error variable

$$\tilde{v}_x = v_x(t) - v_x^*(t) \, .$$

Then, the equations of motion can be written in the following form

$$\dot{\tilde{v}}_x = -\dot{\theta} v_z \, , \tag{16}$$

$$\dot{v}_z = u_1 + \dot{\theta}(\tilde{v}_x + v_x^*(t)) \, , \tag{17}$$

$$\ddot{\theta} = u_2 \, , \tag{18}$$

$$\ddot{\psi} = -u_1 c \cos \psi - u_2 - d \sin \psi - e\dot{\psi} \, , \tag{19}$$

where

$$c = \frac{m_f a}{I_f + m_f a^2}, \, d = \frac{Fc}{m + m_f}, \, e = \frac{\epsilon}{I_f + m_f a^2} \, ,$$

and $(u_1, \, u_2)$ are new control inputs defined as in [4].

Now, we consider the following candidate Lyapunov function for the system (16)-(19):

$$V = \frac{r_1}{2}(\tilde{v}_x^2 + v_z^2) + \frac{r_2}{2}\theta^2 + \frac{r_3}{2}\dot{\theta}^2 + r_4 d(1 - \cos \psi) + \frac{r_4}{2}(\dot{\theta} + \dot{\psi})^2 \tag{20}$$

where r_1, r_2, r_3, and r_4 are positive constants. The function V is positive definite in the domain

$$D = \{(\tilde{v}_x, \, v_z, \, \theta, \, \dot{\theta}, \, \psi, \, \dot{\psi}) \mid -\pi < \psi < \pi\} \, .$$

The time derivative of V along the trajectories of (16)-(19) is

$$\dot{V} = [r_1 v_z - r_4 c(\dot{\theta} + \dot{\psi}) \cos \psi] u_1 - r_4 e(\dot{\theta} + \dot{\psi})^2 + [r_1 v_x^*(t) v_z + r_2 \theta + r_3 u_2 + r_4 e(\dot{\theta} + \dot{\psi}) - r_4 d \sin \psi]\dot{\theta} \, .$$

Clearly, the feedback laws

$$u_1 = -l_1[r_1 v_z - r_4 c(\dot{\theta} + \dot{\psi}) \cos \psi] \, , \tag{21}$$

$$u_2 = -l_2\dot{\theta} - [\frac{r_1}{r_3} v_x^*(t) v_z + \frac{r_2}{r_3}\theta] - \frac{r_4}{r_3}[e(\dot{\theta} + \dot{\psi}) - d \sin \psi)] \tag{22}$$

where l_1 and l_2 are positive constants, yield

$$\dot{V} = -l_1[r_1 v_z - r_4 c(\dot{\theta} + \dot{\psi}) \cos \psi]^2 - l_2 \dot{\theta}^2 - r_4 e(\dot{\theta} + \dot{\psi})^2 \, ,$$

which satisfies $\dot{V} \le 0$ in D. Using LaSalle's principle, it is easy to prove asymptotic stability of the origin of the closed loop defined by the equations (16)-(19) and the feedback control laws (21)-(22). Note that the positive gains r_i, $i = 1, \, 2, \, 3, \, 4$ and l_j, $j = 1, \, 2$, can be chosen arbitrarily to achieve good closed loop responses.

IV. Simulation

In this section, we demonstrate the effectiveness of the controller by performing simulations of the closed loop defined by the nonlinear equation (4)-(7) and the feedback control laws (21)-(22). The physical parameters used in the simulation are $m = 600 \, kg$, $I = 720 \, kg/m^2$, $m_f = 100 \, kg$, $I_f = 90 \, kg/m^2$, $a = 0.32 \, m$, $b = 0.25 \, m$, $F = 500 \, N$ and $\epsilon = 0.19 \, kg \cdot m^2/s$ [4]. We consider stabilization of the spacecraft in orbital transfer, suppressing the transverse and pitching motion of the spacecraft and sloshing of fuel while the spacecraft is accelerating. In other words, the control objective is to stabilize the relative equilibrium corresponding to a constant axial spacecraft acceleration of $0.714 \, m/s^2$ and $v_z = \theta = \dot{\theta} = \psi = \dot{\psi} = 0$.

Time responses are shown in Figures 3 and 4 correspond to the initial conditions $v_{x0} = 10000 \, m/s$, $v_{z0} = 350 \, m/s$, $\theta_0 = 2^o$, $\dot{\theta}_0 = 0.57^o/s$, $\psi_0 = 30^o$, and $\dot{\psi}_0 = 0$.

The transverse velocity, attitude angle and the slosh angle converge to the relative equilibrium at zero while the axial velocity v_x increases and $\dot{v}_x$ tends asymptotically to $0.714 \, m/s^2$. Note that there is trade-off between good responses for the directly actuated degrees of freedom (the transverse and pitch dynamics) and good responses for the unactuated degree of freedom (the slosh dynamics); the controller given by (21)-(22) with gain values $r_1 = 10^{-7}$, $r_2 = 10$, $r_3 = 10^2$, $r_4 = 10^{-2}$, $l_1 = 10^3$, $l_2 = 10^2$ represent one example of this balance.

V. Conclusions

We have developed dynamic models for a spacecraft with a single fuel sloshing mode, considering both pendulum and mass-spring analogies. These models can be extended in a number of ways; for example, it is possible to develop multi-mass models which allow representation of higher frequency slosh modes.

We have designed a nonlinear feedback controller which achieves suppression of the slosh mode, as well as the transverse and pitch dynamics, while the spacecraft accelerates in axial direction. We have illustrated the effectiveness of the controller through a simulation example.

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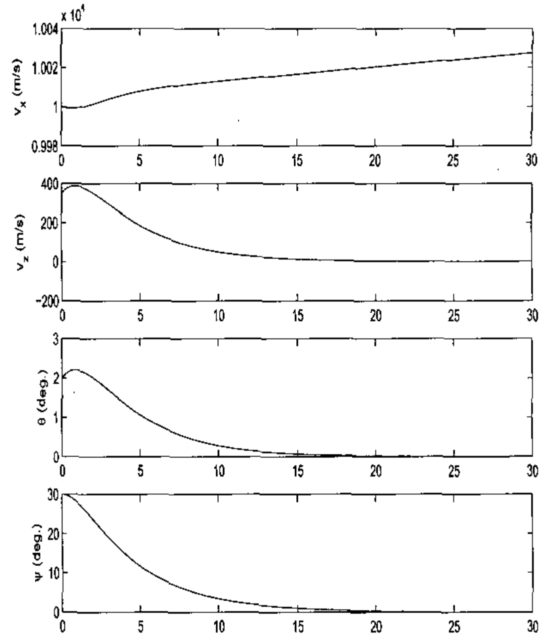

Fig. 3. State variables v_x, v_z, θ, and ψ.

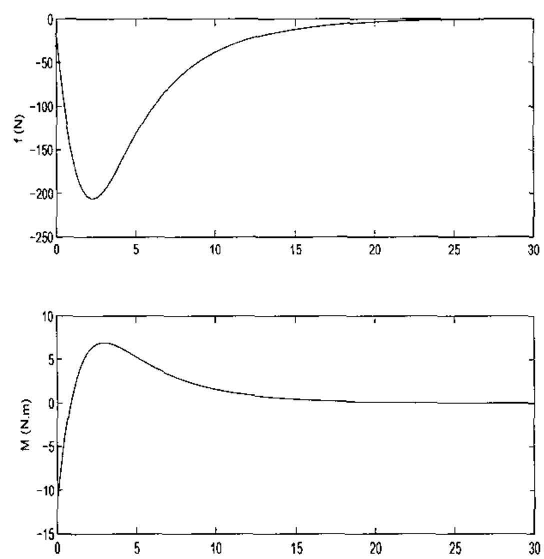

Fig. 4. Transverse control force f and pitching moment M.